

Discrimination and Calibration Decouple in Runtime Failure Monitors Under Distribution Shift

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Abstract—Runtime monitors estimate failure risk while a robot policy executes, but the monitor is exposed to the same distribution shift as the policy it supervises. Across 1,746 episodes from two vision language action policies and seven perturbation dimensions in LIBERO-Plus, we evaluate five monitoring signals and find that distribution shift can substantially alter calibration even when failure discrimination remains informative. A novelty based monitor, for example, retains useful ranking while its empirical coverage falls to 0.157 against a nominal target of 0.90 and it alarms on most episodes that the robot successfully completes. We introduce DISMON, a feature based monitor that predicts episode failure from internal policy representations while reducing sensitivity to nuisance variation. On perturbation categories excluded from monitor training, combining DISMON with action uncertainty improves AUROC by 0.054 and 0.052 over the stronger individual signal on the two policies without requiring additional policy forward passes beyond those already used for uncertainty estimation. The benefit of nuisance suppression decreases with perturbation severity. Recalibration restores coverage to 0.770 using unlabeled target episodes and to 0.899 using delayed episode outcomes. These results show that failure discrimination alone does not characterize monitoring under distribution shift, since deployment also depends on whether the operating threshold remains calibrated.

Index Terms—robot learning, runtime monitoring, distribution shift, vision language action models, calibration

I. INTRODUCTION

Vision language action (VLA) policies that perform well under their training conditions can degrade when deployed under changes in camera viewpoint, appearance, object arrangement, language, or robot state. Benchmarks such as LIBERO-Plus [1] and LIBERO-PRO [2] make these effects measurable through controlled perturbations of the operating environment. Runtime monitoring provides a complementary mechanism for managing such failures. A monitor observes execution and estimates failure risk so that a supervisory system can intervene when necessary. Existing methods construct monitoring signals from internal representations [3], action uncertainty and novelty [4], action representations [5], task success prediction [6], world model features [7], temporal consistency [8], and activation perturbations [9].

The monitor, however, is exposed to the same distribution shift as the policy it supervises. It may continue to assign larger scores to failed episodes while the overall score distribution changes enough that a threshold selected before deployment no longer behaves as intended. AUROC can therefore remain

informative because score ordering is preserved even while the fixed threshold begins rejecting successful executions. This distinction is particularly important for novelty based monitors because an unfamiliar observation does not necessarily imply that the policy cannot complete the task.

Most runtime monitoring methods are evaluated under conditions close to those used for development or calibration, a limitation also noted by SAFE [3]. Robustness benchmarks instead perturb the policy without examining whether the monitor remains calibrated under the same shift. Conformal methods can adapt predictive thresholds under distribution shift [10], [11], including in planning and runtime assurance [12]–[15], but have not been studied for failure monitors receiving only delayed episode outcomes. We therefore evaluate several monitoring signals on identical shifted episodes and report empirical coverage alongside AUROC.

The monitoring signals respond to shift in different ways. Novelty remains associated with failure while producing alarms on most successful shifted episodes, whereas action uncertainty remains substantially better calibrated. We introduce DISMON, a lightweight monitor trained on features from the policy’s ordinary acting pass to preserve outcome predictive information while reducing sensitivity to nuisance variation. Camera viewpoint, object layout, initial state, and language are excluded from monitor training and reserved for transfer evaluation. Neither DISMON nor uncertainty consistently dominates when used alone, but their combination improves AUROC on both policies without increasing the policy forward-pass cost already required for uncertainty estimation. We further examine when nuisance suppression improves transfer and whether calibration can be recovered after deployment.

II. METHOD

A. Runtime Monitoring Under Distribution Shift

At control step t , a monitor produces a scalar score s_t , where larger values indicate greater evidence of failure. Because intervention can occur as soon as the score crosses a threshold, we summarize each episode by

$$S = \max_t s_t, \quad (1)$$

which provides a common episode level score for methods that otherwise use different policy outputs. We evaluate five

signals: action entropy across stochastic samples [4], a supervised feature probe on internal activations [3], random network distillation (RND) novelty [4], action temporal consistency across overlapping predictions [8], and activation perturbation [9].

Failure discrimination is measured by AUROC with failure treated as the positive class and stratified bootstrap intervals. Comparisons on paired scores use bootstrap resampling over identical episodes and are cross checked with DeLong’s test. Calibration is measured through empirical coverage under a threshold estimated from successful reference episodes. Given n calibration scores, the split conformal threshold is

$$\hat{q} = S_{(\lceil (n+1)(1-\alpha) \rceil)}, \quad (2)$$

with $1-\alpha = 0.90$. Coverage is the fraction of successful target episodes satisfying $S \leq \hat{q}$. We also report detection lead time, defined as the number of steps between the first alarm and the end of a failed episode.

AUROC depends on score ordering within the evaluation distribution, whereas coverage depends on scores relative to the fixed reference threshold. A shift can therefore preserve ranking while pushing successful episodes beyond $\hat{q}$.

B. DISMON

A forward hook records a pooled hidden state $f_t \in \mathbb{R}^D$ during the policy’s acting pass. The trajectory prefix through step t is summarized causally by

$$u_t = \left[\frac{1}{t+1} \sum_{i \leq t} f_i ; \max_{i \leq t} f_i \right], \quad (3)$$

allowing the monitor to score each prefix during execution. An MLP maps u_t to z_t with a binary failure prediction head. Prefixes are supervised using the final episode outcome because supervision only at the complete episode creates a mismatch with the max-over-steps score used at runtime.

Internal policy representations can encode environmental variation unrelated to outcome, allowing a monitor trained directly on these features to identify the perturbation rather than failure. We therefore train DISMON using

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \lambda_c \mathcal{L}_{\text{con}} + \lambda_n \mathcal{L}_{\text{nui}}, \quad (4)$$

where the contrastive term encourages representations with the same episode outcome to remain similar across nuisance conditions. The nuisance classifier is connected through gradient reversal, with its weight gradually increased so that nuisance suppression begins after the failure predictor starts to form. Development uses clean, lighting, texture, and sensor noise. Camera viewpoint, object layout, initial state, and language are excluded entirely, so evaluation measures transfer across perturbation categories. Every DISMON result uses a three seed ensemble.

C. Fusion, Cost, and Recalibration

Entropy measures uncertainty in the policy’s current action prediction, whereas DISMON learns representations associated with the eventual episode outcome. These signals need not coincide because a policy may be uncertain and recover, or fail without a corresponding rise in uncertainty. We therefore

evaluate a fitted fusion model trained only on development conditions and a parameter free average of standardized scores. The fitted model uses cross fitting so that each training episode receives a DISMON score from a monitor that was not trained on that episode.

Monitoring cost is measured in additional policy forward passes. Entropy estimated from K samples requires $K - 1$ additional passes, while activation perturbation requires M . The supervised probe, RND, and DISMON reuse features from the acting pass and require no additional policy evaluations.

For adaptation after deployment, weighted conformal calibration uses unlabeled target episodes [10]. Adaptive conformal inference updates the threshold using the delayed binary outcome of each rollout [11], without identifying the step at which failure occurs.

III. EXPERIMENTAL SETUP

Experiments use LIBERO-Plus [1], which extends LIBERO [16] with controlled perturbations sampled deterministically across benchmark conditions together with a reference calibration cell. $\pi_{0.5}$ [17] is an action chunking flow matching VLA that replans every ten steps, with $K = 5$ and $M = 8$. OpenVLA [18] is autoregressive and single step, with $K = 3$, $M = 4$, and sampled signals refreshed every ten steps. We use vanilla OpenVLA rather than OpenVLA-OFT [19], whose regression action head would make sampled entropy degenerate. Temporal consistency is undefined for OpenVLA because no future timestep is predicted more than once. Monitor training draws comparable numbers of prefixes per episode from both policies despite their different control cadences.

The dataset contains 1,746 episodes: 1,120 for $\pi_{0.5}$ with 148 failures and 626 for OpenVLA with 473 failures. After removing calibration and clean probe training splits, 1,036 and 542 episodes respectively remain for evaluation. As a check against an integration error, the same OpenVLA adapter completes five of six original LIBERO `libero_spatial` episodes under the official protocol, consistent with the published success rate of approximately 85%. Code, split manifests, seeds, and rollout traces are released.

Table I reports success by perturbation dimension. Because the stratified sample places greater weight on the three DISMON training dimensions, the aggregate rates are not directly comparable with benchmark averages. The held out conditions are harder, with $\pi_{0.5}$ falling to 55.0% success under initial-state shift and OpenVLA to 0.0% and 1.7% under camera and initial-state shifts.

IV. RESULTS

A. Calibration Under Distribution Shift

Table II reports pooled AUROC and coverage for the five monitoring signals. RND shows the clearest divergence between discrimination and calibration. On $\pi_{0.5}$, it achieves an AUROC of 0.542 while empirical coverage falls to 0.157 from the nominal target of 0.90. On OpenVLA, RND provides stronger discrimination with an AUROC of 0.697, but coverage

TABLE I
POLICY SUCCESS RATE PER SHIFT DIMENSION (EVALUATION SPLIT).
AGGREGATES ARE WEIGHTED TOWARD THE TRAINING DIMENSIONS BY
DESIGN.

Dimension	$\pi_{0.5}$		OpenVLA	
	n	succ.	n	succ.
clean (calib. remainder)	36	100.0%	36	41.7%
lighting [§]	200	97.0%	90	15.6%
background texture [§]	200	96.5%	108	38.0%
sensor noise [§]	200	84.5%	70	11.4%
language	100	88.0%	60	40.0%
object layout	100	77.0%	58	29.3%
camera	100	76.0%	60	0.0%
initial state	100	55.0%	60	1.7%
all	1036	85.7%	542	22.1%

[§]DISMON training dimension; the four below the rule are held out.

TABLE II
MONITOR SIGNALS UNDER POOLED SHIFT. COVERAGE IS THAT OF THE
CLEAN-CALIBRATED THRESHOLD AT NOMINAL 0.90.

Signal	$\pi_{0.5}$ (1,120 ep.)		OpenVLA (626 ep.)	
	AUROC	Cov.	AUROC	Cov.
Action-chunk entropy	0.812	0.865	0.822	0.961
Activation perturbation	0.737	0.939	0.755	0.913
RND OOD score	0.542	0.157	0.697	0.292
Action temporal consist.	0.700	0.902	0.500 [‡]	1.000
Supervised feature probe	n/a [†]	0.901	0.509	1.000

[†]Untrainable: zero failures in the probe-training split. [‡]Undefined by construction on a single-step policy.

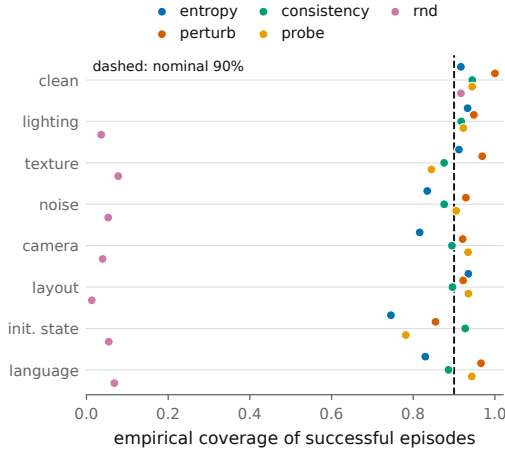


Fig. 1. Empirical coverage per shift dimension, nominal 0.90. RND holds 0.157 on $\pi_{0.5}$; entropy stays near nominal.

remains only 0.292. Action entropy stays much closer to nominal coverage while also providing the strongest discrimination among the individual signals on both policies. Figure 1 shows that the RND coverage loss is distributed across perturbation dimensions rather than being determined by one condition.

Table III separates the two principal error patterns. RND flags 95.1% of successful shifted episodes while missing only 2.8% of failures, so its low miss rate is accompanied by an extremely high false alarm rate. The supervised probe exhibits

TABLE III
ERROR DECOMPOSITION ON SHIFTED DIMENSIONS ($\pi_{0.5}$), AT THE
CLEAN-CALIBRATED THRESHOLD.

Signal	False alarm	Miss	Lead time (steps)
RND OOD score	0.951	0.028	27.9*
Supervised probe	0.105	0.955	25.6*
Activation perturbation	0.070	0.714	15.2
Action-chunk entropy	0.142	0.354	15.0
Action temporal consist.	0.104	0.768	12.6

*Not meaningful for a signal alarming on nearly every episode (RND) or nearly none (probe); the probe’s 0.105/0.955 corner is the degenerate behaviour a coverage-only report would disguise as healthy.

the opposite behavior, flagging only 10.5% of successful episodes but missing 95.5% of failures. Its high empirical coverage therefore reflects infrequent alarms rather than effective failure detection. Entropy provides a more balanced operating point, with a 14.2% false alarm rate and a 35.4% miss rate.

Detection lead time must also be interpreted together with these errors. RND produces the earliest apparent warning at 27.9 steps, but this value is difficult to interpret when the monitor alarms on nearly every shifted episode.

The probe also exposes a limitation of clean-data supervision. The $\pi_{0.5}$ training split contains no failures, while the OpenVLA probe fits its clean split at 0.963 AUROC but transfers at only 0.509. This motivates training monitors on perturbed rather than exclusively clean rollouts.

B. DISMON Complements Action Uncertainty

On the four held out perturbation categories, neither DISMON nor entropy consistently dominates. Entropy performs better on $\pi_{0.5}$, with AUROC 0.846 compared with 0.827 for DISMON, whereas DISMON performs better on OpenVLA, reaching 0.875 compared with 0.835 for entropy.

Combining the signals improves performance on both policies. The parameter free fusion reaches 0.900 AUROC on $\pi_{0.5}$, an improvement of 0.054 with a 95% confidence interval of $[+0.022, +0.089]$, $p = 0.0002$, and DeLong $p = 0.0016$. On OpenVLA, it reaches 0.887, improving by 0.052 with a confidence interval of $[+0.014, +0.095]$ and $p = 0.0054$. A signed Stouffer combination across the disjoint evaluation sets gives $p = 4.3 \times 10^{-6}$.

The fitted combiner produces nearly the same result, indicating that the improvement does not depend on fitting a more flexible fusion rule. On the $\pi_{0.5}$ language condition, for example, DISMON reaches 0.716 AUROC and entropy reaches 0.884, while their fusion reaches 0.947.

Figure 2 shows the corresponding accuracy and monitoring cost tradeoff. When repeated action sampling is available, adding DISMON to entropy improves AUROC without increasing the forward-pass cost already required by entropy. When this sampling budget is unavailable, DISMON provides a zero additional pass alternative.

C. When Nuisance Suppression Helps

We evaluate three DISMON variants trained on identical data. The adversarial rung improves held out AUROC by

TABLE IV
HELD-OUT-DIMENSION AUROC AND MONITORING COST, IN EXTRA
POLICY FORWARD PASSES PER CONTROL STEP. THE FUSION AND ENTROPY
COST THE SAME.

Monitor	$\pi_{0.5}$	OpenVLA	Extra fwd.
Fusion (parameter-free)	0.900	0.887	$K-1$
Fusion (fitted)	0.899	0.889	$K-1$
Entropy	0.846	0.835	$K-1$
DISMON-C	0.827	0.875	0
Activation perturbation	0.794	—	M
Probe / RND	≈ 0.50	≈ 0.50	0

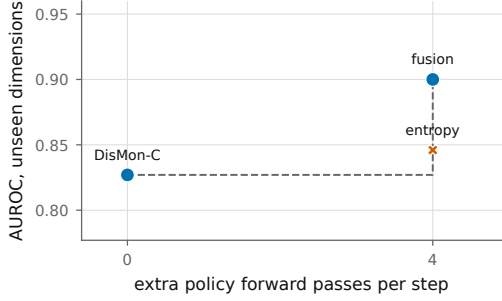


Fig. 2. Accuracy against monitoring cost on $\pi_{0.5}$. DISMON holds the zero-cost corner; the fusion dominates entropy at identical cost.

TABLE V
DISMON RUNGS ON FOUR HELD-OUT DIMENSIONS ($\pi_{0.5}$), 3-SEED
ENSEMBLES; ALL RUNGS SEE IDENTICAL DATA AND DIFFER ONLY IN
OBJECTIVE.

Rung	Objective	Pooled	Per-seed	Δ vs. A
A	augmentation	0.792	.795/.790/.786	—
B	+ contrastive	0.789	.795/.760/.804	−0.004
C	+ adversarial	0.827	.804/.780/.835	+0.034*

* $p = 0.0060$, Holm-corrected. Rung C per dimension: camera 0.776, layout 0.874, initial state 0.807, language 0.757.

0.034 over the matched augmentation baseline, with a 95% confidence interval of $[+0.010, +0.059]$ and Holm-corrected $p = 0.0060$. All four held out dimensions show improvement in the same direction.

The ablation in Table V attributes the gain to the adversarial term rather than contrastive alignment alone. Adding contrastive alignment changes AUROC by -0.004 relative to the augmentation baseline with $p = 0.573$, whereas adding gradient reversal to the contrastive objective improves AUROC by 0.038 with $p = 0.0004$. This separates the effect of training on perturbed data from the effect of the nuisance suppression objective.

The result is conditional on the three trained ensembles. Bootstrap resampling captures variation across evaluation episodes rather than independent monitor training runs, and rung C is less stable across seeds than rung A. We therefore interpret the result as evidence for these trained ensembles rather than a general claim that adversarial nuisance suppression will always improve monitoring.

We further stratify the same $\pi_{0.5}$ evaluation by LIBERO-Plus perturbation severity while holding the policy, checkpoint, features, and episode length fixed. The advantage of rung C

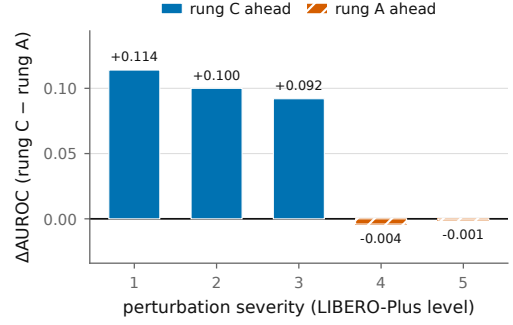


Fig. 3. Rung-C advantage over the matched-data baseline by perturbation severity, within $\pi_{0.5}$. The benefit decays as the operating point hardens.

TABLE VI
EMPIRICAL COVERAGE UNDER THE CALIBRATION LADDER ($\pi_{0.5}$,
NOMINAL 0.90), ORDERED BY FEEDBACK REQUIRED.

Rung	Feedback required	Baseline	DISMON-C
L0 fixed	none (frozen)	0.895	0.598
L1 weighted	unlabeled episodes	0.934	0.770
L2 ACI	delayed outcomes	0.900	0.899
L3 $k=5$	5 labels	0.930	0.810
L3 $k=10$	10 labels	0.823	0.936
L3 $k=25$	25 labels	0.914	0.906

L3 rows are a single draw of which k episodes were labeled and are non-monotone in k ; conclusions are drawn from L1 and L2 only.

over the matched baseline decreases from $+0.114$ AUROC at severity levels 1–2 to -0.001 at levels 4–5. The prespecified low versus high severity contrast is -0.113 , with a 95% confidence interval of $[-0.173, -0.056]$ and $p = 0.0002$.

The result suggests that nuisance suppression helps while the shifted variation remains largely unrelated to outcome, but loses its advantage once that variation itself becomes predictive of failure.

D. Calibration After Deployment

A fixed reference threshold on the DISMON-C score produces only 0.598 empirical coverage under distribution shift. Weighted conformal calibration, which requires only unlabeled target episodes, raises coverage to 0.770. Adaptive conformal inference using delayed binary episode outcomes reaches 0.899, close to the nominal target of 0.90. Figure 4 shows the corresponding variation across perturbation dimensions.

Adaptive conformal inference requires only the final episode outcome, not the timestep at which failure occurred. We also report the few shot calibration results but do not draw a conclusion from them because coverage is nonmonotonic as the number of labeled episodes increases.

Statistical protocol. Rung comparisons use the stratified paired bootstrap with Holm correction and are conditional on the three trained ensembles. Our initial analysis compared four per-dimension aggregates using a sign-flip permutation test, whose minimum attainable two-sided p -value at $n = 4$ is $2/2^4 = 0.125$. We therefore adopted the episode level analysis before computing results under it, with the protocol fixed

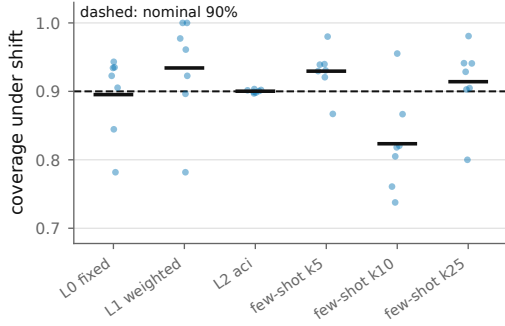


Fig. 4. Coverage per calibration rung with per-dimension spread; the dashed line is nominal 0.90, the vertical axis truncated to the realised range.

at a public commit on 2026-08-02, and retain the aggregate analysis as a secondary check.

V. LIMITATIONS

The experiments are simulation-only and cover two policies with a single episode-collection seed, so bootstrap uncertainty does not capture collection or retraining variance. The DISMON ablation is conditional on three trained ensembles, which is insufficient to fully separate objective effects from training instability. LIBERO-Plus provides no completely unperturbed calibration condition, and several OpenVLA shifts contain too few successful episodes for reliable calibration analysis. Finally, the experiments establish complementarity between DISMON and entropy rather than optimality of the pair, while the severity result indicates that nuisance suppression should not be assumed to help uniformly across operating conditions.

VI. CONCLUSION

We study runtime monitoring under the same distribution shifts encountered by the policy being supervised. Across 1,746 episodes, monitoring signals retain different amounts of failure information while exhibiting substantially different threshold behavior. RND provides the clearest example, with empirical coverage falling to 0.157 against a nominal target of 0.90 and a 95.1% false alarm rate on successful shifted episodes.

DISMON predicts episode failure from internal policy representations while reducing sensitivity to nuisance variation. Its predictions complement action uncertainty, and their fusion improves held out AUROC on both policies without additional policy forward passes beyond those required for entropy. The ablation attributes the transfer improvement to adversarial nuisance suppression, while the severity analysis shows that its benefit decreases as the shifted variable becomes more predictive of failure.

Finally, unlabeled target episodes raise DISMON coverage from 0.598 to 0.770, while delayed binary outcomes raise it to 0.899. Reliable runtime monitoring under distribution shift therefore requires both an informative failure signal and an operating threshold appropriate for the deployment distribution. Code, split manifests, seeds, and rollout traces are released.

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